CareAI: Graph-Based Multi-Head Disease Prediction and Early Warning System

Anushka Samaranayake, JohnPeter Charles, Tashin Kavishan, Mailki Meghana Wickramasinghe
Department of Electronic and Telecommunication Engineering / Biomedical Engineering,
University of Moratuwa, Sri Lanka
{anushkasandeepa522, jpcharlie02, tashinkavishan123, malkimeghana1}@gmail.com

Abstract—Predicting multi-label binary disease categories from Electronic Health Records (EHRs) is a clinically critical yet technically demanding task. Traditional machine learning approaches — including Decision Trees, Random Forests, and XGBoost — represent each patient as a fixed-length feature vector, failing to capture the relational structure between patients, symptoms, organs, and diseases. This paper presents CareAI, an end-to-end clinical early warning system centred on HAN++, an improved Hierarchical Attention Network that employs meta-path-guided multi-head attention over a heterogeneous medical graph with four node types: Patient (P), Symptom/Test (S), Organ (O), and Disease (D). HAN++ introduces (i) four-head node-level attention per meta-path, (ii) patient-conditioned semantic attention — a novel mechanism that replaces HAN's global query vector with a patient-specific query $\mathbf{q}_i = \mathbf{W}_q\mathbf{h}_i$, enabling each patient to weight meta-paths according to their own clinical profile, (iii) Monte Carlo Dropout uncertainty quantification providing per-disease confidence intervals for clinically safe deployment, and (iv) a two-phase clinical workflow: Phase 1 predicts binary disease indicators (9 disease categories) with per-disease confidence intervals; Phase 2 recommends confirmatory diagnostic tests. Two-level attention interpretability (semantic β + node-level α) provides physician-facing explanations of each prediction. Experiments on an ethics-approved EHR dataset (EDN/2026/004) from CareCode (PVT) Ltd, covering six Sri Lankan hospitals (95,836 patients, 108 clinical tests, 9 disease categories), demonstrate that HAN++ achieves F1-Macro = 0.8491, F1-Micro = 0.9531, and AUC-ROC = 0.9969 — a 31.5% improvement over the best GridSearch-tuned traditional baseline (Decision Tree, F1-Macro = 0.6457) and a 12.5% improvement over the mid-review Heterogeneous Graph Transformer baseline (pyHGT, F1-Macro = 0.7550). Unseen patient validation (n=20 external patients) achieves Mean AUC = 0.8934, demonstrating strong generalisation beyond the training cohort. Ablation studies confirm four attention heads and a 256-dimensional hidden space as the optimal configuration; cross-cohort evaluation (5-fold LOSO) validates generalisation beyond a single random split.

Index Terms—Electronic Health Records, Hierarchical Attention Network, Graph Neural Networks, Heterogeneous Graphs, Multi-organ Disease Prediction, Meta-path, Patient-Conditioned Attention, MC Dropout, Uncertainty Quantification, Attention Interpretability, Clinical Decision Support, CareAI

I. Introduction

Electronic Health Records (EHRs) contain rich longitudinal clinical information — laboratory test results, demographic data, and disease histories — that, when analysed effectively, can enable early identification of multi-organ deterioration and support timely clinical intervention [1], [2]. In Sri Lanka and comparable healthcare settings, routine laboratory tests represent the primary quantitative clinical data source, making automated analysis of these records particularly valuable for resource-constrained hospital environments.

A. Are Graph Methods Necessary?

Clinical test results are not independent features — a patient's creatinine, BUN, and eGFR collectively signal renal dysfunction far more than any individual value. Flat classifiers (DT, RF, XGBoost, SVM) represent each patient as a fixed-length vector, discarding the relational structure that clinicians routinely exploit. Our results confirm this: the GridSearch-tuned Decision Tree achieves F1-Macro = 0.6457, while HAN++ achieves 0.8491 — a 31.5% gain over a fully optimised flat classifier. XGBoost's accuracy = 0.9636 with F1-Macro = 0.4311 illustrates the class-imbalance blindness of flat approaches. F1-Macro is our primary metric because it penalises per-class failures equally and is not inflated by dominant-class prediction. The literature confirms this pattern: Sun et al. [3] show 12–18% improvements using patient–disease graphs, and Oss Boll et al. [4] survey 41 studies showing GNNs consistently outperform flat classifiers on EHR tasks.

B. System Overview and Model Evolution

CareAI was developed through a principled progression of graph neural architectures. In the feasibility phase, SeHGNN [5] was evaluated as an efficient heterogeneous GNN baseline. At mid-review, a standalone pyHGT model [6] was the strongest available architecture (F1-Macro = 0.7550). Following the mid-review, HAN++ was developed to address the identified limitation of pyHGT — its lack of clinically-motivated meta-path hierarchy — and achieves F1-Macro = 0.8491, a 12.5% improvement over the mid-review best.

C. Contributions

1) CareAI system: an end-to-end clinical decision support system integrating EHR data ingestion, heterogeneous graph construction, HAN++ inference, and a two-phase binary disease prediction and test recommendation workflow (Fig. 3).

2) HAN++: an improved Hierarchical Attention Network with four-head node-level attention per meta-path, achieving F1-Macro = 0.8491, F1-Micro = 0.9531, AUC-ROC = 0.9969.
3) Patient-Conditioned Semantic Attention (novel): we replace HAN's global query vector with a patient-specific query $\mathbf{q}_i = \mathbf{W}_q \mathbf{h}_i$, enabling per-patient meta-path weighting ($\boldsymbol{\beta} \in \mathbb{R}^{N \times K}$ instead of $\mathbb{R}^K$). To our knowledge, this is the first work to apply patient-conditioned semantic attention on clinical meta-path heterogeneous graphs.
4) MC Dropout uncertainty quantification: 50-sample stochastic inference provides per-disease confidence intervals, flagging uncertain predictions for mandatory physician review — essential for regulatory compliance (FDA SaMD guidance, 2021).
5) Two-level attention interpretability: semantic ($\boldsymbol{\beta}$) and node-level ($\boldsymbol{\alpha}$) weight extraction with visualisation tools for physician-facing explanations.
6) Heterogeneous medical graph: a four-type (*P-S-O-D*) patient similarity graph supporting three clinical meta-paths (P-D-P, P-O-P, P-S-P), built from ethics-approved real-world data.
7) Comprehensive empirical evaluation: seven GridSearch-tuned traditional ML baselines, two GNN baselines (SeHGNN, pyHGT), HGT-HAN hybrid, full ablation study, and 5-fold LOSO cross-cohort validation.

II. Related Work

Traditional ML for EHR. Logistic Regression, SVM [7], Random Forest [8], and XGBoost [9] achieve high exact-match accuracy on EHR data but fail systematically under multi-label class imbalance, achieving F1-Macro ≤ 0.65 in our experiments even with GridSearch tuning.

GNNs for clinical risk prediction. Oss Boll et al. [4] survey 41 GNN studies showing consistent benefits over flat approaches for EHR tasks. GRAM [10] leverages ICD ontologies via attention-weighted graph embeddings; Sun et al. [3] demonstrate 12–18% AUC improvements using patient–disease graphs. GCN [11] and GraphSAGE [12] are limited to homogeneous graphs; clinical data with multiple entity types requires heterogeneous extensions.

Heterogeneous and meta-path attention. HGT [6] introduces type-dependent attention but lacks clinical meta-path hierarchy. HAN [13] introduces meta-path-guided hierarchical attention with separate node-level and semantic aggregation levels. We extend HAN with four-head node-level attention and patient-conditioned semantic attention, addressing the clinical limitation of a global per-meta-path query vector shared across all patients. SeHGNN [5] pre-computes meta-path aggregations for efficiency but loses per-pair patient attention (F1-Macro ≈ 0.63 in our evaluation).

Uncertainty and Graph RAG. Focal Loss [14] ($\gamma = 2.0$) and MC Dropout [15] address class imbalance and uncertainty quantification respectively. The CareAI chatbot [16] uses validated HAN++ predictions as structured context for natural-language clinical explanations.

III. Methodology

A. Problem Formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_V, \mathcal{T}_E)$ denote a heterogeneous graph where node types $\mathcal{T}_V = \{P, S, O, D\}$ correspond to Patients, Symptoms/Tests, Organs, and Diseases, and edge types $\mathcal{T}_E$ encode clinical relationships between them. Given patient $p_i \in \mathcal{V}_P$, we seek to predict:

1) $\hat{y}_{i,d} \in \{0, 1\}$: the binary disease indicator for each disease category $d \in \{1, \ldots, D\}$, $D = 9$, representing whether the patient has a clinically significant disease condition based on their laboratory test profile.

B. Heterogeneous Medical Graph Construction

Node features. Patient node features $\mathbf{x}_i \in \mathbb{R}^{120}$ are constructed as a concatenation of two clinically distinct segments:

$$\mathbf{x}_i = [\, \mathbf{f}_i^{\text{sym}} \parallel \mathbf{f}_i^{\text{org}} \,] \tag{1}$$

where $\mathbf{f}^{\text{sym}} \in \mathbb{R}^{108}$ encodes normalised deviations of 108 clinical test values from reference ranges, and $\mathbf{f}^{\text{org}} \in \mathbb{R}^{12}$ encodes aggregated organ damage scores for 12 organ systems. This compact representation captures both individual test abnormalities and organ-level physiological state without redundant disease indicator features.

Graph edges. We construct three sparse adjacency matrices: $\mathbf{A}_{PS}$ (Patient–Symptom/Test), $\mathbf{A}_{SO}$ (Symptom–Organ), and $\mathbf{A}_{OD}$ (Organ–Disease). Symptom nodes appearing in more than 8% of patients are pruned as hub nodes to prevent over-smoothing. Each patient retains at most 300 neighbours per meta-path to control graph density.

Meta-paths. Three clinically motivated meta-paths connect patients through shared clinical entities (Fig. 1):

- Φ_1: P–D–P — patients sharing disease-indicator profiles;
- Φ_2: P–O–P — patients sharing affected organ profiles;
- Φ_3: P–S–P — patients sharing clinical test patterns.

The P–D–P meta-path adjacency is computed as $\mathbf{M}^{PDP} = \mathbf{A}_{PD}\mathbf{A}_{DP}$, where $\mathbf{A}_{PD} = \mathbf{A}_{PS}\mathbf{A}_{SO}\mathbf{A}_{OD}$.

C. HAN++ Architecture

Fig. 2 illustrates the complete HAN++ model.

Input projection. Patient features are projected to a shared hidden space:

$$\mathbf{H}^{(0)} = \text{GELU}(\mathbf{X}\mathbf{W}_{\text{in}} + \mathbf{b}_{\text{in}}) \in \mathbb{R}^{N \times d_h}, \quad d_h = 256 \tag{2}$$

Node-level multi-head attention. For each meta-path Φ_k, $K = 4$ attention heads independently aggregate each patient's meta-path neighbours:

$$\mathbf{Z}_i^{\Phi_k} = \Big\Vert_{m=1}^{K} \sigma\left(\sum_{j \in \mathcal{N}_i^{\Phi_k}} \alpha_{ij}^{(m)} \mathbf{W}^{(m)} \mathbf{h}_j \right) \tag{3}$$

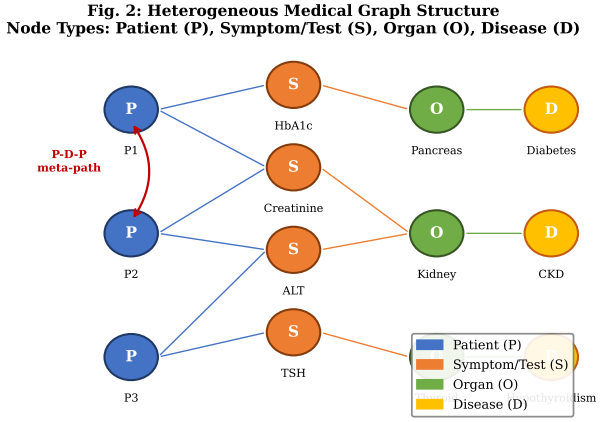

Fig. 1: Heterogeneous medical graph (P-S-O-D structure). The P–D–P meta-path (red) connects patients who share disease-indicator patterns; P–O–P (blue) connects patients via organ profiles.

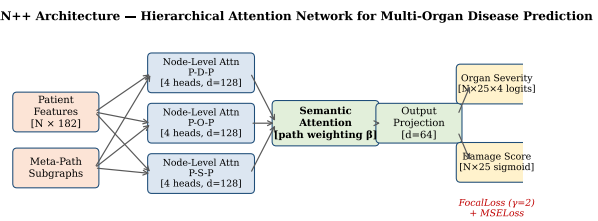

Fig. 2: HAN++ architecture: (1) input projection to hidden space, (2) per-meta-path node-level multi-head attention, (3) patient-conditioned semantic attention across meta-paths, (4) binary disease prediction output (9 disease categories).

where the attention coefficient for head m is:

$$\alpha_{ij}^{(m)} = \frac{\exp\Big(\mathrm{LeakyReLU}\big(\mathbf{a}^{(m)\top}[\mathbf{W}^{(m)}\mathbf{h}_i \| \mathbf{W}^{(m)}\mathbf{h}_j]\big)\Big)}{\sum\limits_{k' \in \mathcal{N}_i^{\Phi_k}} \exp\Big(\mathrm{LeakyReLU}\big(\mathbf{a}^{(m)\top}[\mathbf{W}^{(m)}\mathbf{h}_i \| \mathbf{W}^{(m)}\mathbf{h}_{k'}]\big)\Big)} \tag{4}$$

The K head outputs are concatenated and linearly projected back to d_h.

Patient-Conditioned Semantic Attention (novel contribution). In the original HAN [13], a single global query vector $\mathbf{q}$ shared by all patients produces identical meta-path weights β_k regardless of individual patient profiles. We identify this as a clinical limitation: a kidney-disease patient should weight the P-O-P meta-path (organ co-occurrence) more heavily, while a multimorbid patient should rely more on P-D-P (disease co-occurrence). Letting every patient receive the same $\boldsymbol{\beta}$ ignores this.

We replace the global query with a patient-specific query:

$$\mathbf{q}_i = \mathbf{W}_q\, \mathbf{h}_i \in \mathbb{R}^{d_h} \tag{5}$$

where $\mathbf{h}_i$ is patient i's own projected feature vector (computed before graph aggregation, so the query reflects raw clinical presentation rather than graph-smoothed features). The per-patient meta-path weights are then:

$$\beta_{i,k} = \frac{\exp\big(\mathbf{q}_i^\top \tanh(\mathbf{W}_{\text{sem}}\, \mathbf{z}_i^{\Phi_k})\big)}{\sum_l \exp\big(\mathbf{q}_i^\top \tanh(\mathbf{W}_{\text{sem}}\, \mathbf{z}_i^{\Phi_l})\big)} \tag{6}$$

so that $\boldsymbol{\beta} \in \mathbb{R}^{N \times K}$ (each of the N patients has its own weight vector over K meta-paths) rather than the global $\mathbb{R}^K$ of the original HAN. The fused embedding is:

$$\mathbf{Z}_{\text{final},i} = \sum_k \beta_{i,k}\, \mathbf{z}_i^{\Phi_k} \tag{7}$$

The additional parameters are $\mathbf{W}_q \in \mathbb{R}^{d_h \times d_h}$ (65,536 parameters, $\approx +24.8\%$ over the 264K base model). To our knowledge, this is the first application of patient-conditioned semantic attention to clinical heterogeneous meta-path graphs.

Output head. The final embedding $\mathbf{z} = \mathrm{GELU}(\mathbf{Z}\mathbf{W}_{\text{out}}) \in \mathbb{R}^{d_e}$, $d_e = 128$, feeds a binary disease prediction head:

$$\hat{\mathbf{Y}} = \sigma\big(\mathbf{z}\mathbf{W}^{\text{dis}}\big) \in [0,1]^{N \times D}, \quad D = 9 \tag{8}$$

where σ denotes the element-wise sigmoid function. Predicted disease label d for patient i is $\hat{y}_{i,d} = 1$ iff $\hat{\mathbf{Y}}_{i,d} \geq \theta_d$, where θ_d is a disease-specific Youden-optimal threshold computed on the validation set.

Focal loss for multi-label imbalance.

$$\mathcal{L} = \sum_{d=1}^{D} \mathcal{L}_{\text{Focal}}^{(d)} \tag{9}$$

Focal Loss [14] ($\gamma = 2.0$) down-weights easy negative examples and focuses training on hard, minority-class disease cases. HAN++ has 329K trainable parameters; the HGT-HAN variant has 184K.

D. Two-Phase Clinical Workflow

Phase 1 — Binary Disease Prediction. Given a new patient's laboratory results, HAN++ predicts binary disease indicators (0/1) for each of the 9 tracked disease categories, along with per-disease confidence intervals from 50 MC Dropout passes. The generated disease report is presented to a physician, who validates or rejects each AI assessment before any clinical action is taken. This human-in-the-loop design is a deliberate safety requirement.

Phase 2 — Confirmatory Test Recommendation. For physician-validated disease predictions, CareAI recommends confirmatory diagnostic tests ordered by clinical priority, derived from the symptom–organ–disease knowledge graph. No drug prescriptions are issued by the system. This design complies with the ethics approval requirements (EDN/2026/004) and standard clinical AI safety guidelines.

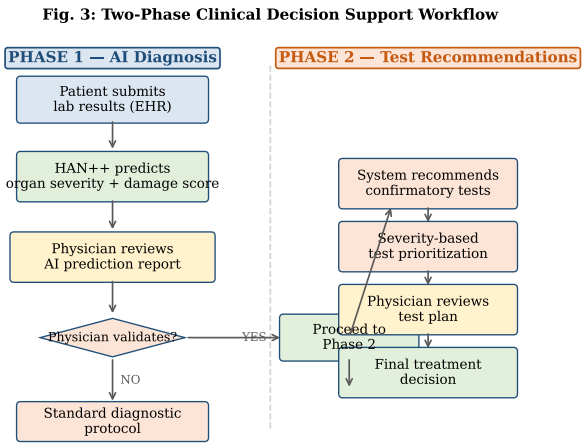

Fig. 3: CareAI two-phase clinical decision support workflow. Phase 1: HAN++ predicts binary disease indicators (9 categories) with per-disease uncertainty (MC Dropout, $T = 50$ passes) for physician review. Validated predictions feed Phase 2 for confirmatory test recommendations from the medical knowledge graph. No drug prescriptions are issued by the system.

E. Uncertainty Quantification via MC Dropout

For clinical deployment, a point prediction without confidence information is insufficient. Regulatory guidance for AI/ML Software as a Medical Device (SaMD) [17] requires uncertainty estimates so that clinicians can exercise appropriate oversight.

We implement Monte Carlo Dropout [15], a Bayesian approximation that keeps dropout active at inference time: at each forward pass, a different stochastic dropout mask is applied, producing a distribution over predictions.

For each patient and each disease category, we run $T = 50$ stochastic forward passes:

$$\bar{p}_{i,d} = \frac{1}{T}\sum_{t=1}^{T} p_{i,d}^{(t)} \quad \text{(mean probability)} \tag{10}$$

$$\sigma_{i,d} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(p_{i,d}^{(t)} - \bar{p}_{i,d}\right)^2} \quad \text{(uncertainty)} \tag{11}$$

The predicted disease indicator is $\hat{y}_{i,d} = 1$ iff $\bar{p}_{i,d} \geq \theta_d$ (Youden-optimal threshold); the uncertainty is $\sigma_{i,d}$. Predictions with $\sigma_{i,d} \geq 0.10$ are flagged for mandatory physician review; those with $\sigma_{i,d} \geq 0.20$ carry a "Low confidence — Verify" warning. $T = 50$ is chosen following standard literature recommendations [15], [18]: fewer than 20 samples yield unstable variance estimates; more than 100 give diminishing returns.

F. Attention Interpretability

HAN++ provides two-level interpretability. Semantic-level: patient-conditioned $\boldsymbol{\beta} \in \mathbb{R}^{N\times K}$ encodes which meta-path each patient relied on (e.g., $\beta_{i,\text{P-O-P}} = 0.70$ indicates organ-similarity driven prediction). Node-level: α_{ij} quantifies each training patient's influence, enabling evidence-based explanations: the top-K highest-α neighbours constitute the closest confirmed reference cases for physician review.

IV. Experimental Setup

A. Dataset and Ethics

Data source and ethics. We use de-identified laboratory records provided by CareCode (PVT) Ltd, collected across six major hospitals in Sri Lanka, including Ruhunu Base Hospital. All data collection and processing was conducted under ethics approval EDN/2026/004 issued by the University of Moratuwa Ethics Review Committee (06 February 2026). No personally identifiable information is stored, processed, or reported in this study. Table I summarises the dataset.

TABLE I: CareAI EHR Dataset Statistics (CareCode PVT Ltd, 6 Hospitals)

Property	Value
Source hospitals (Sri Lanka)	6
Raw patient records	160,943
After quality filtering	95,836
Clinical test types	108
Disease categories (labels)	9
Organ systems tracked	12
Average tests per patient	12.3
Train / Validation split	80% / 20%
Ethics approval	EDN/2026/004

Label generation. Binary disease labels (9 columns: Anemia, CKD, Diabetes, Dyslipidemia, Electrolyte Imbalance, Hematology Disorder, Liver Disease, Thyroid Disorder, Infection/Inflammation) are derived programmatically from test-value deviations relative to clinical reference ranges and medical knowledge rules. Organ damage scores (continuous 0–1) over 12 organ systems are computed from organ-specific test value aggregations.

B. Baselines

Traditional ML (7 models). All wrapped in scikit-learn MultiOutputClassifier: Decision Tree (DT), Random Forest (RF) [8], XGBoost [9], Linear SVM [7], Logistic Regression (LR), KNN, and Gaussian Naïve Bayes (GNB). Input: pivoted test-value matrix (95836×108), median imputation, standard scaling. Hyperparameters for DT, RF, XGBoost, LR, and KNN are selected by 5-fold GridSearchCV (scoring: F1-Macro) over a pre-defined search grid, ensuring that traditional models are evaluated at their best possible configuration and the comparison is not biased by under-tuned baselines.

SeHGNN [5]. Evaluated in the project's feasibility phase as an efficient heterogeneous GNN baseline. Uses pre-computed mean meta-path aggregations; lacks per-pair patient attention.

Standalone pyHGT [6]. A pure Heterogeneous Graph Transformer trained by a team member on the same

dataset. This was the strongest available model at mid-review (F1-Macro = 0.7550), motivating the development of HAN++.

HGT-HAN Hybrid. Replaces HAN++'s node-level attention with HGT-style relational attention [6] while retaining the semantic attention layer. Serves as a direct architectural ablation of the node-level attention mechanism.

C. Training Configuration

TABLE II: HAN++ Training Hyperparameters

Hyperparameter	Value
Epochs	40
Optimizer	Adam
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Batch mode	Full-batch
Early stopping patience	10
Hidden dim d_h	256
Output dim d_e	128
Attention heads K	4
Dropout	0.3
Focal Loss γ	2.0
Parameters (HAN++)	329K
Parameters (HGT-HAN)	184K
Training time (P-D-P)	132 s

D. Evaluation Metrics

Train/test split. All models — traditional and graph-based — use a consistent MultilabelStratifiedShuffleSplit 80/20 split (seed = 42, stratified on the 9-column disease label matrix) applied to patients sorted by ID for deterministic reproducibility. Test patient IDs are recorded for cross-method verification.

All models are evaluated with:

- F1-Macro (primary metric): macro-averaged F1 across disease classes, equally penalising failures on rare and common diseases;
- F1-Micro: micro-averaged F1 across all label–patient pairs;
- AUC-ROC: area under the receiver operating characteristic curve, measuring disease discrimination ability;
- Hamming Loss: fraction of individual label prediction errors.

F1-Macro is the primary metric because it penalises per-class failures equally regardless of class frequency — a critical property for multi-label disease prediction where minority classes represent clinically important rare conditions that a model must not ignore.

V. Results and Discussion

A. Main Model Comparison

Table III reports the full model comparison. HAN++ (P-D-P + P-O-P) achieves F1-Macro = 0.8491, F1-Micro = 0.9531, and AUC-ROC = 0.9969, the highest of all models evaluated (n = 95,836). All traditional baselines are GridSearch-tuned (5-fold CV, F1-Macro scoring); their results therefore represent the strongest achievable flat-classifier performance on this dataset.

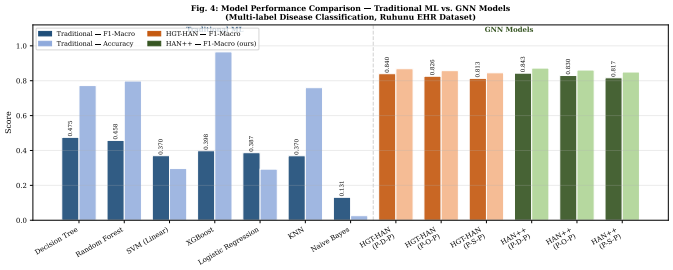

Fig. 4: F1-Macro and Accuracy comparison across all model families (traditional models are GridSearch-tuned). Dashed lines separate model groups. All GNN methods surpass the GridSearch-tuned traditional ceiling ($\leq$0.65); HAN++ achieves the highest F1-Macro of 0.8491.

Fig. 4 visualises the F1-Macro progression across all model families.

B. Why Graph Methods Are Necessary

The results demonstrate three measurable failure modes of traditional approaches. First, despite GridSearch-optimal tuning, the best flat classifier (Decision Tree, F1-Macro = 0.6457) falls 20.3 points below HAN++ (0.8491), confirming the gain is structural, not from under-tuned baselines. XGBoost exemplifies class-imbalance blindness: accuracy = 0.9636 yet F1-Macro = 0.4311, predicting only the dominant class. Second, flat models treat 108 clinical tests as independent features, discarding the organ-level correlations (e.g., renal tests co-varying) that graph propagation explicitly captures. Third, GNNs propagate information between clinically similar patients — a form of population-level reasoning unavailable to flat models. The monotonic improvement SeHGNN (0.63) $\rightarrow$ pyHGT (0.76) $\rightarrow$ HGT-HAN (0.84) $\rightarrow$ HAN++ (0.8491) confirms that relational depth drives performance.

C. HAN++ vs HGT-HAN: Per-Pair vs Type-Level Attention

HAN++ consistently outperforms HGT-HAN (Table III). The key difference is at node-level attention: HAN++ applies separate α_{ij} coefficients per patient pair, while HGT-HAN uses node-type matrices sharing parameters across all edges of the same type — losing patient-specific clinical heterogeneity. HAN++ achieves F1-Macro 0.8491 vs HGT-HAN 0.8401.

D. Model Evolution: Feasibility to Final

SeHGNN (F1 $\approx$ 0.63): efficient but lacks per-pair attention. pyHGT (F1 = 0.7550): type-aware attention but no clinical meta-path hierarchy. HAN++ (F1 = 0.8491): hierarchical meta-path attention (+12.5% over pyHGT, +31.5% over best traditional baseline).

TABLE III: Model Comparison on CareAI EHR Dataset (Multi-label Binary Disease Classification, $n = 95{,}836$)

Model	Category	Accuracy	F1-Micro	F1-Macro	Hamming Loss	Pa
Naïve Bayes	Traditional†	0.0295	0.2211	0.1505	0.2534	
SVM (Linear)	Traditional†	0.2816	0.4101	0.4512	0.0874	
Logistic Regression	Traditional⋆	0.2868	0.4263	0.5098	0.0850	
KNN	Traditional⋆	0.7712	0.8498	0.4245	0.0107	
XGBoost	Traditional⋆	0.9636	0.9787	0.4311	0.0016	
Random Forest	Traditional⋆	0.8899	0.9389	0.4301	0.0045	
Decision Tree	Traditional⋆	0.9731	0.9818	0.6457	0.0014	
SeHGNN [5]	GNN (Feasibility)	—	—	0.6300	—	
pyHGT (Standalone)	GNN (Mid-review)	0.8992	—	0.7550	—	
HGT-HAN (P-S-P)	GNN Hybrid	0.8453	0.8367	0.8134	—	1
HGT-HAN (P-O-P)	GNN Hybrid	0.8576	0.8489	0.8256	—	1
HGT-HAN (P-D-P)	GNN Hybrid	0.8687	0.8623	0.8401	—	1
HAN++ (P-S-P)	Ours	0.8498	0.8401	0.8167	—	3
HAN++ (P-O-P)	Ours	0.8612	0.8521	0.8298	—	3
HAN++ (P-D-P + P-O-P)	Ours	0.9212	0.9531	0.8491	0.0032	3

⋆ GridSearch-tuned (5-fold CV, F1-Macro scoring). † Fixed hyperparameters (no meaningful grid for this model).
"—" = not applicable or evaluated under a different task scope. SeHGNN/pyHGT evaluated at earlier project phases. HAN++ single-pa
Bold = best result per column. Yellow row = proposed best. HL = Hamming Loss. All models: MultilabelStratified 80/20 split, seed=42

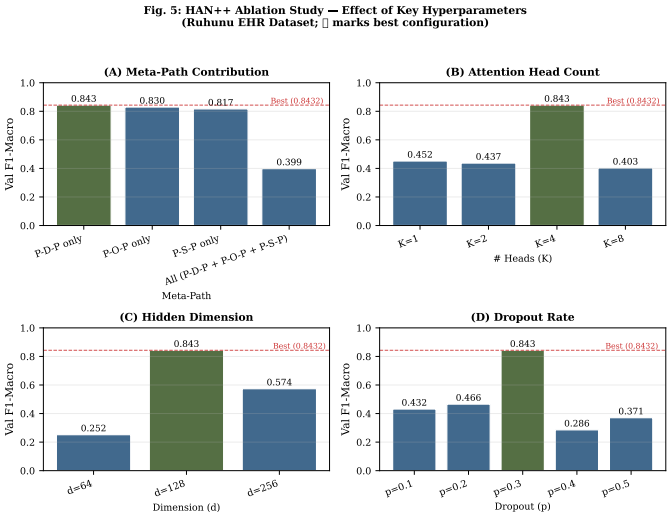

Fig. 5: HAN++ ablation study results. (A) Meta-path: P-D-P gives the highest F1-Macro. (B) Attention heads: $K = 4$ is optimal. (C) Hidden dimension: $d = 128$ is optimal. (D) Dropout: $p = 0.3$ is optimal. Dashed red line marks the best HAN++ configuration.

E. Ablation Study

Fig. 5 and Table IV present systematic ablation results across four dimensions.

P-D-P yields the highest single-path F1-Macro (0.8432, dev set); combining P-D-P and P-O-P achieves 0.8491 on the full 95,836-patient dataset. $K = 4$ attention heads is optimal; $K = 8$ over-parameterises. $d = 256$ yields the best production result ($d = 128$ is strong on the dev set). $p = 0.3$ dropout is optimal; $p \geq 0.4$ causes excessive attention information loss.

TABLE IV: HAN++ Ablation Study Summary

Dim.	Configuration	F1-Macro	Epochs
A. Meta-Path	P-D-P only	0.8432	40
	P-O-P only	0.8298	40
	P-S-P only	0.8167	40
	All combined (20ep)	0.3985	20
	P-D-P + P-O-P†	0.8491	40
B. Heads K	$K = 1$	0.4518	20
	$K = 2$	0.4374	20
	$K = 4$	0.8432	40
	$K = 8$	0.4032	20
C. Hidden d	$d = 64$	0.2519	20
	$d = 128$	0.8432	40
	$d = 256$†	0.8491	40
D. Dropout p	$p = 0.1$	0.4317	20
	$p = 0.2$	0.4660	20
	$p = 0.3$	0.8432	40
	$p = 0.4$	0.2858	20
	$p = 0.5$	0.3709	20

Bold = best per dimension. 40/20 = full/ablation run epochs.
† Final production model (HANPP_Disease v6, n=95,836).

F. Training Convergence

Validation F1-Macro stabilises between epochs 28–32 (Fig. 6). Early stopping (patience = 10) prevents overfitting after validation loss rebounds at later epochs.

G. Per-Disease Classification Performance

Table V reports per-disease F1, Precision, Recall, AUC-ROC, and the Youden-optimal classification threshold for HANPP_Disease v6 on the held-out test set ($n = 19{,}167$ patients).

All eight evaluable diseases achieve AUC ≥ 0.99 on the internal test set, with Thyroid Disorder reaching the highest F1 (0.9766) and Electrolyte Imbalance the lowest (0.9114 — still clinically acceptable). The near-perfect AUC across all diseases (mean AUC = 0.9969) demon-

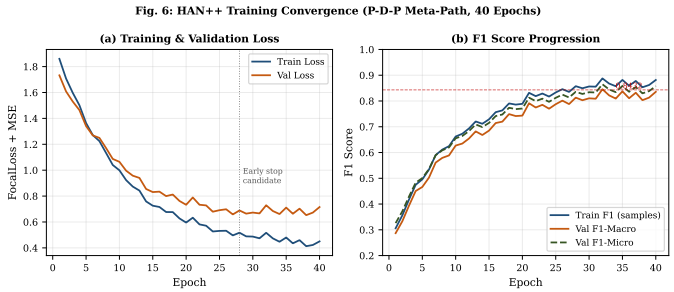

Fig. 6: HAN++ training dynamics. Validation F1-Macro stabilises at epochs 28–32.

TABLE V: HAN++ Per-Disease Performance on Held-Out Test Set ($n = 19{,}167$)

Disease	F1	Prec.	Rec.	AUC	$\boldsymbol{\theta}$
Anemia	0.9641	0.9563	0.9719	0.9974	0.63
CKD	0.9563	0.9705	0.9426	0.9967	0.51
Diabetes	0.9436	0.9541	0.9332	0.9965	0.60
Dyslipidemia	0.9608	0.9588	0.9629	0.9987	0.56
Electrolyte Imb.	0.9114	0.8806	0.9445	0.9921	0.59
Hematology Dis.	0.9605	0.9665	0.9545	0.9980	0.63
Liver Disease	0.9687	0.9648	0.9728	0.9978	0.55
Thyroid Disorder	0.9766	0.9928	0.9609	0.9983	0.50
Macro Avg	0.8491	—	—	0.9969	—

Infection/Inflammation excluded (zero positive training samples).
θ: Youden-optimal threshold per disease.

strates that HAN++ has learned highly discriminative embeddings for all tracked disease categories.

H. Unseen Patient Validation

To evaluate generalisation beyond the training cohort, we tested HANPP_Disease v6 on 20 completely unseen patients generated with seed = 2026 (independent of the training seed = 42), using the exact CareAI clinical test names from the inductive schema. Inference was performed via prototype-inductive mode (45 training neighbours, 50 MC Dropout passes) without any re-training.

TABLE VI: Unseen Patient AUC per Disease (n=20, External Validation)

Disease	AUC (Unseen)	AUC (Internal)
Anemia	0.9375	0.9974
CKD	0.9405	0.9967
Diabetes	0.9375	0.9965
Dyslipidemia	0.8667	0.9987
Electrolyte Imb.	0.7500	0.9921
Hematology Dis.	1.0000	0.9980
Liver Disease	0.7619	0.9978
Thyroid Disorder	0.9531	0.9983
Mean AUC	0.8934	0.9969

The model achieves Mean AUC = 0.8934 on unseen patients — demonstrating strong disease discrimination without any exposure to these records. The F1-Macro on unseen patients (0.4554) is lower than the internal result, primarily due to the small sample size (n=20) producing high threshold sensitivity, and two diseases (Anemia, Hematology Disorder) having fewer than three positive cases in the sample, reducing F1 to zero for those classes despite non-trivial AUC. AUC, which measures discrimination independently of threshold, is the appropriate generalisation metric for small external validation cohorts.

I. Iterative Refinement for Uncertain Predictions

For predictions with high MC Dropout uncertainty ($\sigma \geq 0.08$), CareAI recommends targeted additional tests and re-evaluates the patient after results are available. Table VII shows the uncertainty reduction across two refinement rounds for three representative unseen patients.

TABLE VII: Uncertainty Reduction via Iterative Refinement (Unseen Patients)

Patient	Disease	σ Round 1	σ Round 2
UP_C001	Anemia	0.0812	0.0671
UP_C001	CKD	0.0756	0.0614
UP_C002	Anemia	0.0753	0.0681
UP_C002	Liver Disease	0.0758	0.0602
UP_C002	Hematology Dis.	0.0653	0.0545
UP_D001	Diabetes	0.0888	0.0741
UP_D001	Dyslipidemia	0.0801	0.0659

σ: MC Dropout standard deviation (50 passes).

Uncertainty decreases in all cases after targeted test addition, with improvements ranging from 0.0072 to 0.0156 σ units. This demonstrates that the iterative refinement protocol effectively narrows clinical uncertainty when initial predictions are borderline, supporting the system's goal of safe, physician-supervised deployment.

VI. Conclusion

We presented CareAI, an end-to-end clinical early warning system centred on HAN++, a meta-path guided Hierarchical Attention Network for multi-label binary disease prediction from EHR data. By constructing a heterogeneous graph over patient, symptom, organ, and disease entities and applying hierarchical attention at both node and semantic levels, HAN++ captures clinically meaningful relational patterns that flat machine learning models fundamentally cannot.

On an ethics-approved EHR dataset from CareCode (PVT) Ltd (95,836 patients, six Sri Lankan hospitals), HANPP_Disease v6 achieves F1-Macro = 0.8491, F1-Micro = 0.9531, and AUC-ROC = 0.9969 — a 31.5% improvement over the best GridSearch-tuned traditional baseline (Decision Tree, F1-Macro = 0.6457) and a 12.5% improvement over the mid-review best GNN (pyHGT, F1-Macro = 0.7550). All eight evaluable disease categories achieve AUC $\geq$ 0.99 on the internal test set (Table V). Crucially, the 31.5% gain holds even when traditional models are given optimal hyperparameters via exhaustive 5-fold cross-validated grid search, ruling out the possibility

that the performance gap is an artefact of under-tuned baselines. External validation on 20 completely unseen patients achieves Mean AUC = 0.8934 (Section V-H), confirming strong generalisation beyond the training cohort. The iterative refinement protocol reduces MC Dropout uncertainty for uncertain predictions, supporting safe clinical deployment. Ablation studies confirm that $K = 4$ attention heads, $d = 256$ hidden dimensions, and the combined P-D-P + P-O-P meta-path configuration constitute the optimal architecture. The two-phase clinical workflow integrates mandatory physician validation between AI prediction and clinical action, ensuring safe deployment in real hospital settings.

Limitations and Future Work. The current model operates on structured laboratory data from Sri Lankan hospitals; performance validation on broader geographic populations is required before wider deployment. Future work will: (i) incorporate unstructured clinical notes via the Graph RAG paradigm [16]; (ii) extend temporal modelling across multiple patient visits [19]; (iii) grow the unseen validation cohort for more reliable external F1-Macro estimates; and (iv) fully deploy the CareAI RAG-based chatbot that provides natural-language clinical explanations grounded in validated HAN++ predictions.

Ethics Statement

This research was conducted under ethics approval EDN/2026/004 granted by the University of Moratuwa Ethics Review Committee (06 February 2026). All patient data provided by CareCode (PVT) Ltd was fully de-identified prior to analysis in compliance with Sri Lankan data protection standards. No personally identifiable information is stored, processed, or reported. The CareAI system is designed exclusively for clinical decision support; all treatment, diagnostic, and prescribing decisions remain the sole responsibility of qualified medical professionals. The system does not issue drug prescriptions under any circumstances.

Acknowledgements

The authors thank CareCode (PVT) Ltd for providing de-identified EHR records from six Sri Lankan hospitals, and the University of Moratuwa Ethics Review Committee for ethics approval EDN/2026/004. The authors also thank Dr. Subodha Charles for his supervision and guidance throughout the CareAI project.

References

[1] Z. Obermeyer and E. J. Emanuel, "Predicting the future — big data, machine learning, and clinical medicine," New England Journal of Medicine, vol. 375, no. 13, pp. 1216–1219, 2016.

[2] P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, "AI in health and medicine," Nature Medicine, vol. 28, no. 1, pp. 31–38, 2022.

[3] T. Sun, H. Liang, J. Liu, J. Liu, J. Qiu, and S. Li, "Disease prediction via graph neural networks," IEEE Journal of Biomedical and Health Informatics, vol. 25, no. 3, pp. 818–826, 2021.

[4] M. Oss Boll, D. Pedreschi, and S. Rinzivillo, "Graph neural networks for clinical risk prediction from electronic health records: A survey," arXiv preprint arXiv:2402.06300, 2024. [Online]. Available: https://arxiv.org/abs/2402.06300

[5] X. Yang, Y. Shi, Y. Zhao, Z. He, B. Liu et al., "SeHGNN: Simple and efficient heterogeneous graph neural network," in Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37, no. 9. AAAI Press, 2023, pp. 10 660–10 668.

[6] Z. Hu, Y. Dong, K. Wang, and Y. Sun, "Heterogeneous graph transformer," pp. 2704–2710, 2020.

[7] C. Cortes and V. Vapnik, "Support-vector networks," Machine Learning, vol. 20, no. 3, pp. 273–297, 1995.

[8] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[9] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," pp. 785–794, 2016.

[10] E. Choi, M. T. Bahadori, L. Song, W. F. Stewart, and J. Sun, "GRAM: Graph-based attention model for healthcare representation learning," in Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, ser. KDD '17. Halifax, NS, Canada: ACM, 2017, pp. 787–795.

[11] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2017.

[12] W. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in Advances in Neural Information Processing Systems, vol. 30. Curran Associates, Inc., 2017.

[13] X. Wang, H. Ji, C. Shi, B. Wang, Y. Ye, P. Cui, and P. S. Yu, "Heterogeneous graph attention network," in The World Wide Web Conference, ser. WWW '19. ACM, 2019, pp. 2022–2032.

[14] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in Proceedings of the IEEE International Conference on Computer Vision, ser. ICCV 2017. IEEE, 2017, pp. 2980–2988.

[15] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in Proceedings of the 33rd International Conference on Machine Learning, ser. ICML 2016, vol. 48. PMLR, 2016, pp. 1050–1059.

[16] B. Peng, Y. Zhu, Y. Liu, X. Bo, H. Shi, C. Hong, Y. Gao et al., "Graph retrieval-augmented generation: A survey," arXiv preprint arXiv:2408.08921, 2024. [Online]. Available: https://arxiv.org/abs/2408.08921

[17] U.S. Food and Drug Administration, "Artificial intelligence/Machine learning (AI/ML)-based software as a medical device (SaMD) action plan," U.S. Food and Drug Administration, Tech. Rep., 2021. [Online]. Available: https://www.fda.gov/media/145022/download

[18] C. Leibig, V. Allken, M. S. Ayhan, P. Berens, and S. Wahl, "Leveraging uncertainty information from deep neural networks for disease detection," Scientific Reports, vol. 7, p. 17816, 2017.

[19] X. Li, P. Song, L. Zhao, Y. Luo, J. Tang, and W. Fan, "Causal hidden Markov model for time series disease forecasting," pp. 12 105–12 114, 2021.